

In [1]:

```
import pandas as pd
import numpy as np
from datetime import datetime
```

In [2]:

```
df = pd.read_csv('churn.csv')
df.head()
```

Out[2]:

	Unnamed: 0	age	gender	security_no	region_category	membership_category	joining_date	joined_th
0	0	18	F	XW0DQ7H	Village	Platinum Membership	2017-08-17	
1	1	32	F	5K0N3X1	City	Premium Membership	2017-08-28	
2	2	44	F	1F2TCL3	Town	No Membership	2016-11-11	
3	3	37	M	VJGJ33N	City	No Membership	2016-10-29	
4	4	31	F	SVZXCWB	City	No Membership	2017-09-12	

5 rows × 24 columns

In [3]:

```
df.columns
```

Out[3]:

```
Index(['Unnamed: 0', 'age', 'gender', 'security_no', 'region_category',
      'membership_category', 'joining_date', 'joined_through_referral',
      'referral_id', 'preferred_offer_types', 'medium_of_operation',
      'internet_option', 'last_visit_time', 'days_since_last_login',
      'avg_time_spent', 'avg_transaction_value', 'avg_frequency_login_days',
      'points_in_wallet', 'used_special_discount',
      'offer_application_preference', 'past_complaint', 'complaint_status',
      'feedback', 'churn_risk_score'],
      dtype='object')
```

In [4]:

```
df = df.drop(columns=[c for c in df.columns if c.startswith("Unnamed")])
```

In [5]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 36992 entries, 0 to 36991
Data columns (total 23 columns):
#   Column                Non-Null Count  Dtype
---  -
0   age                    36992 non-null  int64
1   gender                 36992 non-null  object
2   security_no            36992 non-null  object
```

3	region_category	31564	non-null	object
4	membership_category	36992	non-null	object
5	joining_date	36992	non-null	object
6	joined_through_referral	36992	non-null	object
7	referral_id	36992	non-null	object
8	preferred_offer_types	36704	non-null	object
9	medium_of_operation	36992	non-null	object
10	internet_option	36992	non-null	object
11	last_visit_time	36992	non-null	object
12	days_since_last_login	36992	non-null	int64
13	avg_time_spent	36992	non-null	float64
14	avg_transaction_value	36992	non-null	float64
15	avg_frequency_login_days	36992	non-null	object
16	points_in_wallet	33549	non-null	float64
17	used_special_discount	36992	non-null	object
18	offer_application_preference	36992	non-null	object
19	past_complaint	36992	non-null	object
20	complaint_status	36992	non-null	object
21	feedback	36992	non-null	object
22	churn_risk_score	36992	non-null	int64

dtypes: float64(3), int64(3), object(17)

memory usage: 6.5+ MB

In [6]:

```
df['region_category'].unique()
```

Out[6]:

```
array(['Village', 'City', 'Town', nan], dtype=object)
```

In [7]:

```
df['preferred_offer_types'].unique()
```

Out[7]:

```
array(['Gift Vouchers/Coupons', 'Credit/Debit Card Offers',
      'Without Offers', nan], dtype=object)
```

In [8]:

```
# Standardizing placeholder junk value to proper NaN
placeholder_value = ['?', 'xxxxxxx', 'Unknown', 'unknown', '']
df = df.replace(placeholder_value, np.nan)
```

In [9]:

```
df.info()
```

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 36992 entries, 0 to 36991

Data columns (total 23 columns):

#	Column	Non-Null Count	Dtype
---	-----	-----	-----
0	age	36992 non-null	int64
1	gender	36933 non-null	object
2	security_no	36992 non-null	object
3	region_category	31564 non-null	object
4	membership_category	36992 non-null	object
5	joining_date	36992 non-null	object
6	joined_through_referral	31554 non-null	object
7	referral_id	19146 non-null	object
8	preferred_offer_types	36704 non-null	object
9	medium_of_operation	31599 non-null	object
10	internet_option	36992 non-null	object

```

11 last_visit_time      36992 non-null object
12 days_since_last_login 36992 non-null int64
13 avg_time_spent       36992 non-null float64
14 avg_transaction_value 36992 non-null float64
15 avg_frequency_login_days 36992 non-null object
16 points_in_wallet     33549 non-null float64
17 used_special_discount 36992 non-null object
18 offer_application_preference 36992 non-null object
19 past_complaint        36992 non-null object
20 complaint_status      36992 non-null object
21 feedback              36992 non-null object
22 churn_risk_score      36992 non-null int64

```

dtypes: float64(3), int64(3), object(17)

memory usage: 6.5+ MB

In [10]:

```
df.head()
```

Out[10]:

	age	gender	security_no	region_category	membership_category	joining_date	joined_through_referral
0	18	F	XW0DQ7H	Village	Platinum Membership	2017-08-17	I
1	32	F	5K0N3X1	City	Premium Membership	2017-08-28	Na
2	44	F	1F2TCL3	Town	No Membership	2016-11-11	Y
3	37	M	VJGJ33N	City	No Membership	2016-10-29	Y
4	31	F	SVZXCWB	City	No Membership	2017-09-12	I

5 rows × 23 columns

In [11]:

```
df['avg_frequency_login_days'] = pd.to_numeric(df['avg_frequency_login_days'], errors= 'coerce')
```

In [12]:

```
df['joining_date'] = pd.to_datetime(df['joining_date'], errors = 'coerce')
```

In [13]:

```
df.info()
```

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 36992 entries, 0 to 36991

Data columns (total 23 columns):

#	Column	Non-Null Count	Dtype
0	age	36992 non-null	int64
1	gender	36933 non-null	object
2	security_no	36992 non-null	object
3	region_category	31564 non-null	object
4	membership_category	36992 non-null	object
5	joining_date	36992 non-null	datetime64[ns]
6	joined_through_referral	31554 non-null	object

```

7   referral_id                19146 non-null object
8   preferred_offer_types      36704 non-null object
9   medium_of_operation        31599 non-null object
10  internet_option            36992 non-null object
11  last_visit_time            36992 non-null object
12  days_since_last_login      36992 non-null int64
13  avg_time_spent             36992 non-null float64
14  avg_transaction_value      36992 non-null float64
15  avg_frequency_login_days   33470 non-null float64
16  points_in_wallet           33549 non-null float64
17  used_special_discount      36992 non-null object
18  offer_application_preference 36992 non-null object
19  past_complaint             36992 non-null object
20  complaint_status           36992 non-null object
21  feedback                   36992 non-null object
22  churn_risk_score           36992 non-null int64

```

```

dtypes: datetime64[ns](1), float64(4), int64(3), object(15)
memory usage: 6.5+ MB

```

In [14]:

```

# Handling missing values
df['region_category'] = df['region_category'].fillna('Unknown')

```

In [15]:

```

df['joined_through_referral'] = df['joined_through_referral'].fillna('Not Specified')

```

In [16]:

```

df['preferred_offer_types'] = df['preferred_offer_types'].fillna('No Preference')

```

In [17]:

```

df['avg_frequency_login_days'] = df['avg_frequency_login_days'].fillna(df['avg_frequency_login_days'].median())

```

In [18]:

```

df['points_in_wallet'] = df['points_in_wallet'].fillna(df['points_in_wallet'].median())

```

In [19]:

```

df['medium_of_operation'] = df['medium_of_operation'].fillna('Not Specified')

```

In [20]:

```

df.info()

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 36992 entries, 0 to 36991
Data columns (total 23 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   age                                   36992 non-null  int64
1   gender                               36933 non-null  object
2   security_no                          36992 non-null  object
3   region_category                      36992 non-null  object
4   membership_category                  36992 non-null  object
5   joining_date                         36992 non-null  datetime64[ns]
6   joined_through_referral              36992 non-null  object
7   referral_id                          19146 non-null  object
8   preferred_offer_types                36992 non-null  object
9   medium_of_operation                  36992 non-null  object
10  internet_option                      36992 non-null  object
11  last_visit_time                      36992 non-null  object

```

```

12  days_since_last_login      36992 non-null  int64
13  avg_time_spent             36992 non-null  float64
14  avg_transaction_value      36992 non-null  float64
15  avg_frequency_login_days   36992 non-null  float64
16  points_in_wallet           36992 non-null  float64
17  used_special_discount       36992 non-null  object
18  offer_application_preference 36992 non-null  object
19  past_complaint              36992 non-null  object
20  complaint_status            36992 non-null  object
21  feedback                   36992 non-null  object
22  churn_risk_score            36992 non-null  int64
dtypes: datetime64[ns](1), float64(4), int64(3), object(15)
memory usage: 6.5+ MB

```

In [21]:

```
df.head(20)
```

Out[21]:

	age	gender	security_no	region_category	membership_category	joining_date	joined_through_refe
0	18	F	XW0DQ7H	Village	Platinum Membership	2017-08-17	
1	32	F	5K0N3X1	City	Premium Membership	2017-08-28	Not Speci
2	44	F	1F2TCL3	Town	No Membership	2016-11-11	
3	37	M	VJGJ33N	City	No Membership	2016-10-29	
4	31	F	SVZXCWB	City	No Membership	2017-09-12	
5	13	M	PSG1LGF	City	Gold Membership	2016-01-08	
6	21	M	R3CX1EA	Town	Gold Membership	2015-03-19	
7	42	M	4UJ1551	Unknown	No Membership	2016-07-12	Not Speci
8	44	M	0481QNNQ	Village	Silver Membership	2016-12-14	
9	45	F	ZHP4MCR	Town	No Membership	2016-11-30	
10	25	F	OKZ6GDD	Village	Basic Membership	2016-05-22	
11	60	M	KPI9DD5	City	Gold Membership	2016-01-13	
12	11	M	G2FTZTL	Town	No Membership	2015-01-31	

	age	gender	security_no	region_category	membership_category	joining_date	joined_through_refe
13	34	M	S71G75R	City	Basic Membership	2016-12-07	
14	40	M	QAL9ZY2	City	Premium Membership	2016-11-19	
15	16	F	YTBUEEE	Town	Basic Membership	2017-07-09	Not Speci
16	57	F	RZ6TPTR	City	No Membership	2016-01-27	
17	11	M	3GHN9RC	Town	Silver Membership	2017-01-02	
18	39	M	DYK6E24	Town	No Membership	2017-11-01	
19	45	M	XYB1CDG	City	Premium Membership	2016-02-03	

20 rows × 23 columns

In [22]:

```
# Removing invalid values -- value shouldn't be in negative
for col in ['avg_time_spent', 'avg_transaction_value', 'points_in_wallet']:
    before = (df[col] < 0).sum()
    if before > 0:
        print(f' fixing {before} negavtive values in {col}')
        df[col] = df[col].clip(lower=0)
```

fixing 1719 negavtive values in avg_time_spent
fixing 136 negavtive values in points_in_wallet

In [23]:

```
# age sanity check (dropping obiviously invalid ages if any exist)
df = df[(df['age'] >=10) & (df['age'] <=100)]
```

In [24]:

```
# Feature engineering - business-relevant derived fields
# Tenure in days/months since joining (reference date = latest joining_date in data as p
reference_date = df['joining_date'].max()
df['tenure_days'] = (reference_date - df['joining_date']).dt.days
df['tenure_month'] = (df['tenure_days'] / 30).round(1)
```

In [25]:

```
df.head()
```

Out[25]:

	age	gender	security_no	region_category	membership_category	joining_date	joined_through_referi
0	18	F	XW0DQ7H	Village	Platinum Membership	2017-08-17	I

	age	gender	security_no	region_category	membership_category	joining_date	joined_through_referi
1	32	F	5K0N3X1	City	Premium Membership	2017-08-28	Not Specifi
2	44	F	1F2TCL3	Town	No Membership	2016-11-11	Y
3	37	M	VJGJ33N	City	No Membership	2016-10-29	Y
4	31	F	SVZXCWB	City	No Membership	2017-09-12	I

5 rows × 25 columns

In [26]:

```
# Revenue at risk = transaction value * churn flag
df['revenue_at_risk'] = np.where(
    df['churn_risk_score'] == 1, df['avg_transaction_value'], 0
)
```

In [27]:

```
df.head()
```

Out[27]:

	age	gender	security_no	region_category	membership_category	joining_date	joined_through_referi
0	18	F	XW0DQ7H	Village	Platinum Membership	2017-08-17	I
1	32	F	5K0N3X1	City	Premium Membership	2017-08-28	Not Specifi
2	44	F	1F2TCL3	Town	No Membership	2016-11-11	Y
3	37	M	VJGJ33N	City	No Membership	2016-10-29	Y
4	31	F	SVZXCWB	City	No Membership	2017-09-12	I

5 rows × 26 columns

In [28]:

```
df['avg_frequency_login_days']
```

Out[28]:

```
0      17.0
1      10.0
2      22.0
3       6.0
4      16.0
...
36987   6.0
36988  28.0
36989  16.0
36990  20.0
```

36991 16.0
Name: avg_frequency_login_days, Length: 36992, dtype: float64

In [29]:

```
# Engagement score bucket based on logging frequency
def engagement_bucket(days):
    if pd.isna(days):
        return 'Unknown'
    elif days <= 7:
        return 'Highly Engaged'
    elif days <= 15:
        return 'Moderate Engaged'
    else:
        return 'Low Engaged'

df['engagement_level'] = df['avg_frequency_login_days'].apply(engagement_bucket)
```

In [30]:

```
df.head()
```

Out[30]:

	age	gender	security_no	region_category	membership_category	joining_date	joined_through_refer
0	18	F	XW0DQ7H	Village	Platinum Membership	2017-08-17	I
1	32	F	5K0N3X1	City	Premium Membership	2017-08-28	Not Specifi
2	44	F	1F2TCL3	Town	No Membership	2016-11-11	Y
3	37	M	VJGJ33N	City	No Membership	2016-10-29	Y
4	31	F	SVZXCWB	City	No Membership	2017-09-12	I

5 rows × 27 columns

In [31]:

```
df['membership_category'].unique()
```

Out[31]:

```
array(['Platinum Membership', 'Premium Membership', 'No Membership',
      'Gold Membership', 'Silver Membership', 'Basic Membership'],
      dtype=object)
```

In [32]:

```
# membership tier rank
membership_rank = {
    'No Membership': 0,
    'Basic Membership' : 1,
    'Silver Membership' : 2,
    'Gold Membership' : 3,
    'Premium Membership' : 4,
    'Platinum Membership' : 5
}
```

```
df['membership_rank'] = df['membership_category'].map(membership_rank)
```

In [33]:

```
df.head()
```

Out[33]:

	age	gender	security_no	region_category	membership_category	joining_date	joined_through_referr
0	18	F	XW0DQ7H	Village	Platinum Membership	2017-08-17	I
1	32	F	5K0N3X1	City	Premium Membership	2017-08-28	Not Specifi
2	44	F	1F2TCL3	Town	No Membership	2016-11-11	Y
3	37	M	VJGJ33N	City	No Membership	2016-10-29	Y
4	31	F	SVZXCWB	City	No Membership	2017-09-12	I

5 rows × 28 columns

In [34]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 36992 entries, 0 to 36991
```

```
Data columns (total 28 columns):
```

#	Column	Non-Null Count	Dtype
0	age	36992 non-null	int64
1	gender	36933 non-null	object
2	security_no	36992 non-null	object
3	region_category	36992 non-null	object
4	membership_category	36992 non-null	object
5	joining_date	36992 non-null	datetime64[ns]
6	joined_through_referral	36992 non-null	object
7	referral_id	19146 non-null	object
8	preferred_offer_types	36992 non-null	object
9	medium_of_operation	36992 non-null	object
10	internet_option	36992 non-null	object
11	last_visit_time	36992 non-null	object
12	days_since_last_login	36992 non-null	int64
13	avg_time_spent	36992 non-null	float64
14	avg_transaction_value	36992 non-null	float64
15	avg_frequency_login_days	36992 non-null	float64
16	points_in_wallet	36992 non-null	float64
17	used_special_discount	36992 non-null	object
18	offer_application_preference	36992 non-null	object
19	past_complaint	36992 non-null	object
20	complaint_status	36992 non-null	object
21	feedback	36992 non-null	object
22	churn_risk_score	36992 non-null	int64
23	tenure_days	36992 non-null	int64
24	tenure_month	36992 non-null	float64

```
25 revenue_at_risk          36992 non-null float64
26 engagement_level         36992 non-null object
27 membership_rank          36992 non-null int64
dtypes: datetime64[ns](1), float64(6), int64(5), object(16)
memory usage: 7.9+ MB
```

In [35]:

```
# Complaint flag as boolean
df['has_complaint'] = df['past_complaint'].map({'Yes' : 1, 'No' : 0})
```

In [36]:

```
df.head()
```

Out[36]:

	age	gender	security_no	region_category	membership_category	joining_date	joined_through_refer
0	18	F	XW0DQ7H	Village	Platinum Membership	2017-08-17	I
1	32	F	5K0N3X1	City	Premium Membership	2017-08-28	Not Specifi
2	44	F	1F2TCL3	Town	No Membership	2016-11-11	Y
3	37	M	VJGJ33N	City	No Membership	2016-10-29	Y
4	31	F	SVZXCWB	City	No Membership	2017-09-12	I

5 rows × 29 columns

In [37]:

```
df['feedback'].unique()
```

Out[37]:

```
array(['Products always in Stock', 'Quality Customer Care',
      'Poor Website', 'No reason specified', 'Poor Product Quality',
      'Poor Customer Service', 'Too many ads', 'User Friendly Website',
      'Reasonable Price'], dtype=object)
```

In [38]:

```
# feedback category grouped into positive / negative for quick filtering
negative_feedback = [
    'Poor Website', 'Poor Product Quality', 'Poor Customer Service', 'Too many ads'
]
positive_feedback = [
    'Products always in Stock', 'Quality Customer Care', 'User Friendly Website', 'Reasonable Price'
]

def feedback_sentiment(val):
    if val in negative_feedback:
        return 'Negative'
    elif val in positive_feedback:
        return 'Positive'
    else:
        return 'Neutral'
```

```
df['feedback_sentiment'] = df['feedback'].apply(feedback_sentiment)
```

In [39]:

```
df.head()
```

Out[39]:

	age	gender	security_no	region_category	membership_category	joining_date	joined_through_refer
0	18	F	XW0DQ7H	Village	Platinum Membership	2017-08-17	I
1	32	F	5K0N3X1	City	Premium Membership	2017-08-28	Not Specifi
2	44	F	1F2TCL3	Town	No Membership	2016-11-11	Y
3	37	M	VJGJ33N	City	No Membership	2016-10-29	Y
4	31	F	SVZXCWB	City	No Membership	2017-09-12	I

5 rows × 30 columns

In [40]:

```
len(df)
```

Out[40]:

36992

In [41]:

```
df.info()
```

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 36992 entries, 0 to 36991

Data columns (total 30 columns):

#	Column	Non-Null Count	Dtype
0	age	36992 non-null	int64
1	gender	36933 non-null	object
2	security_no	36992 non-null	object
3	region_category	36992 non-null	object
4	membership_category	36992 non-null	object
5	joining_date	36992 non-null	datetime64[ns]
6	joined_through_referral	36992 non-null	object
7	referral_id	19146 non-null	object
8	preferred_offer_types	36992 non-null	object
9	medium_of_operation	36992 non-null	object
10	internet_option	36992 non-null	object
11	last_visit_time	36992 non-null	object
12	days_since_last_login	36992 non-null	int64
13	avg_time_spent	36992 non-null	float64
14	avg_transaction_value	36992 non-null	float64
15	avg_frequency_login_days	36992 non-null	float64
16	points_in_wallet	36992 non-null	float64
17	used_special_discount	36992 non-null	object
18	offer_application_preference	36992 non-null	object

```
19 past_complaint          36992 non-null object
20 complaint_status        36992 non-null object
21 feedback                 36992 non-null object
22 churn_risk_score         36992 non-null int64
23 tenure_days             36992 non-null int64
24 tenure_month            36992 non-null float64
25 revenue_at_risk          36992 non-null float64
26 engagement_level         36992 non-null object
27 membership_rank          36992 non-null int64
28 has_complaint            36992 non-null int64
29 feedback_sentiment       36992 non-null object
dtypes: datetime64[ns](1), float64(6), int64(6), object(17)
memory usage: 8.5+ MB
```

In [42]:

```
# Dropping rows where critical fields are still missing after cleaning
critical_cols = ['age', 'gender', 'avg_transaction_value', 'churn_risk_score']
len(df)
df = df.dropna(subset=critical_cols)
len(df)
```

Out[42]:

36933

In [43]:

```
# Removiing duplicate rows by security no which act as unique key
df = df.drop_duplicates(subset=['security_no'])
len(df)
```

Out[43]:

36933

In [44]:

```
output_path = 'cleaned_churn.csv'
df.to_csv(output_path, index=False)
```

In [45]:

```
print(f"Final shape: {df.shape}")
print(f"\nChurn rate: {df['churn_risk_score'].mean():.1%}")
print(f"\nColumns: {list(df.columns)}")
```

Final shape: (36933, 30)

Churn rate: 54.1%

Columns: ['age', 'gender', 'security_no', 'region_category', 'membership_category', 'joining_date', 'joined_through_referral', 'referral_id', 'preferred_offer_types', 'medium_of_operation', 'internet_option', 'last_visit_time', 'days_since_last_login', 'avg_time_spent', 'avg_transaction_value', 'avg_frequency_login_days', 'points_in_wallet', 'used_special_discount', 'offer_application_preference', 'past_complaint', 'complaint_status', 'feedback', 'churn_risk_score', 'tenure_days', 'tenure_month', 'revenue_at_risk', 'engagement_level', 'membership_rank', 'has_complaint', 'feedback_sentiment']

In [46]:

```
!pip install sqlalchemy pymysql pandas
```

Collecting sqlalchemy

Downloading sqlalchemy-2.0.52-cp312-cp312-win_amd64.whl.metadata (9.9 kB)

Collecting pymysql

Using cached pymysql-1.2.0-py3-none-any.whl.metadata (4.3 kB)

Requirement already satisfied: pandas in C:\Users\2025a.LAPTOP\anaconda3\envs\myproject

```

\Lib\site-packages (3.0.3)
Collecting greenlet>=1 (from sqlalchemy)
  Downloading greenlet-3.5.5-cp312-cp312-win_amd64.whl.metadata (3.9 kB)
Requirement already satisfied: typing-extensions>=4.6.0 in C:\Users\2025a.LAPTOP\anaconda3\envs\myproject\Lib\site-packages (from sqlalchemy) (4.16.0)
Requirement already satisfied: numpy>=1.26.0 in C:\Users\2025a.LAPTOP\anaconda3\envs\myproject\Lib\site-packages (from pandas) (2.5.1)
Requirement already satisfied: python-dateutil>=2.8.2 in C:\Users\2025a.LAPTOP\anaconda3\envs\myproject\Lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: tzdata in C:\Users\2025a.LAPTOP\anaconda3\envs\myproject\Lib\site-packages (from pandas) (2026.3)
Requirement already satisfied: six>=1.5 in C:\Users\2025a.LAPTOP\anaconda3\envs\myproject\Lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Downloading sqlalchemy-2.0.52-cp312-cp312-win_amd64.whl (2.2 MB)
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----- 2.2/2.2 MB 655.6 kB/s 0:00:03
Using cached pymysql-1.2.0-py3-none-any.whl (45 kB)
Downloading greenlet-3.5.5-cp312-cp312-win_amd64.whl (324 kB)
Installing collected packages: pymysql, greenlet, sqlalchemy

```

Downloading greenlet-3.5.5-cp312-cp312-win_amd64.whl (324 kB)

```
Installing collected packages: pymysql, greenlet, sqlalchemy
```

[illegible]

[illegible]

Successfully installed greenlet-3.5.5 pymysql-1.2.0 sqlalchemy-2.0.52

In [47]:

```
from sqlalchemy import create_engine, text
import pandas as pd

# Update with your actual MySQL credentials
username = "root"
password = "root"
host = "localhost"
port = "3306"
database = "churn_db"

engine = create_engine(f"mysql+pymysql://{username}:{password}@{host}:{port}/{database}")

with engine.connect() as conn:
    print("Connected successfully!")
```

Connected successfully!

In [48]:

```
df.to_sql(
    "customer_churn",
    engine,
    if exists="replace",
```

```
    index=False,  
    chunksize=5000  
)  
print("Data loaded into MySQL successfully!")
```

Data loaded into MySQL successfully!

In []: